University of Ottawa

Department of Mathematics and Statistics

MAT 1341 : Introduction to Linear Algebra
Professor: Hadi Salmasian

First Midterm Exam – Version A

September 25, 2025

Surname _____ First Name _____

Student # _____ DGD _____

Instructions:

(a) You have 80 minutes to complete this exam.
(b) All work to be considered for grading should be written on your solution sheets. You should clearly indicate where the solution of each question is written.
(c) Write your student number at the top of each page in the space provided.
(d) The last page can be detached and be used as scrap paper. You may also use it as extra space for solutions (in this case, do not detach it).
(e) Cellular phones, unauthorized electronic devices or course notes are not allowed during this exam. Phones and devices must be turned off and put away in your bag. Do not keep them in your possession, such as in your pockets. If caught with such a device or document, the following may occur: you will be asked to leave immediately the exam and academic fraud allegations will be filed which may result in you obtaining a 0 (zero) for the exam.
(f) You are strongly recommended to write in **pen**, not pencil.

By signing below, you acknowledge that you are complying with statement (h) above:

Signature: _____

Please do not write in the table below.

Question	1	2	3	4	5	6	Total
Maximum	2	2	4	4	3	4	19
Grade							

1. **[2 points]** Suppose that A is a 4×7 matrix, B is a 6×4 matrix, and C is a 5×8 matrix. For each statement below, write '**T**' if the statement is true, and write '**F**' if the statement is false. You will receive .5 points for each correct answer, lose 0.25 points for each incorrect answer, and receive zero points for an answer left blank. You do not receive a negative score on this question.

____ A linear system with augmented matrix A always has infinitely many solutions.

____ The product AB is a 7×6 matrix.

____ If $\text{rank}(C) = 5$, then for every vector $\mathbf{b} \in \mathbb{R}^5$ the linear system $C\mathbf{x} = \mathbf{b}$ is consistent.

____ If B is nonzero, then the first column of B must be a pivot column.

Solution: F, F, T, F

2. [**2 points**] For each statement below, write '**T**' if the statement is true, and write '**F**' if the statement is false. You will receive .5 points for each correct answer, lose 0.25 points for each incorrect answer, and receive zero points for an answer left blank. You do not receive a negative score on this question.

____ If the cross product of $\mathbf{u}, \mathbf{v} \in \mathbb{R}^3$ vanishes, then $\mathbf{u}$ and $\mathbf{v}$ are parellel to each other.

____ For vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d} \in \mathbb{R}^n$, if $\mathbf{a}$ is a linear combination of $\mathbf{b}, \mathbf{c}$, and $\mathbf{d}$, then $\mathbf{b}$ is a linear combination of $\mathbf{a}, \mathbf{c}$, and $\mathbf{d}$.

____ If A is an $m \times n$ matrix, then for every vector $\mathbf{b} \in \mathbb{R}^n$ the product $A\mathbf{b}$ is a linear combination of the rows of A.

____ For every two vectors $\mathbf{a}, \mathbf{b} \in \mathbb{R}^3$ we have $\mathbf{a} \times (-\mathbf{b}) = \mathbf{b} \times \mathbf{a}$.

Solution: T, F, F, T

3. [**4 points**] Compute the Cartesian equation (i.e., of the form $ax + by + cz = d$) of the plane in $\mathbb{R}^3$ that passes through the point $P = \begin{bmatrix} 0 \\ -1 \\ 3 \end{bmatrix}$ and the line

$$L = \left\{ \begin{bmatrix} 1 \\ -1 \\ 3 \end{bmatrix} + t \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} \ , \ t \in \mathbb{R} \right\}.$$

Hint: First reduce this question to the problem of computing the equation of the plane that passes through 3 given points in $\mathbb{R}^3$.

Solution: For $t = 0$ and $t = 1$ we obtain two points

$$Q = \begin{bmatrix} 1 \\ -1 \\ 3 \end{bmatrix} \quad , \quad R = \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix}$$

On the line L. Thus the plane passes through the points P, Q, R. Now

$$\vec{PQ} = Q - P = \begin{bmatrix} 1 \\ -1 \\ 3 \end{bmatrix} - \begin{bmatrix} 0 \\ -1 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

and

$$\vec{PR} = R - P = \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix} - \begin{bmatrix} 0 \\ -1 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}$$

Thus we obtain a normal vector for the plane as

$$\mathbf{n} = \vec{PQ} \times \vec{PR} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 0 & 0 \\ 2 & 1 & -1 \end{vmatrix} = 0\mathbf{i} - (-1)\mathbf{j} + \mathbf{k} = \mathbf{j} + \mathbf{k} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

Thus if $\mathbf{v}_0 = P$ then

$$\begin{aligned} W &= \{\mathbf{v} \in \mathbb{R}^3 \ : \ (\mathbf{v} - \mathbf{v}_0) \bullet \mathbf{n} = 0\} \\ &= \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3 \ : \ \left(\begin{bmatrix} x \\ y \\ z \end{bmatrix} - \begin{bmatrix} 0 \\ -1 \\ 3 \end{bmatrix} \right) \bullet \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} = 0 \right\} \\ &= \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3 \ : \ 0x + 1(y+1) + 1(z-3) = 0 \right\} \end{aligned}$$

Hence the equation of the plane W is $y + z = 2$.

4. Let $\mathbf{u} = \begin{bmatrix} 1 \\ -1 \\ -2 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$.

(a) [**2 points**] Compute $\text{Proj}_{\mathbf{u}}(\mathbf{v})$.

Solution:

$$\text{Proj}_{\mathbf{u}}(\mathbf{v}) = \frac{\mathbf{u} \bullet \mathbf{v}}{\mathbf{u} \bullet \mathbf{u}}\mathbf{u} = \frac{(1)(1) + (-1)(0) + (-2)(1)}{1^2 + (-1)^2 + (-2)^2}\mathbf{u} = -\frac{1}{6}\mathbf{u} = \begin{bmatrix} -\frac{1}{6} \\ \frac{1}{6} \\ \frac{1}{3} \end{bmatrix}$$

(b) [**2 points**] Is the line $L = \{\mathbf{u} + t\mathbf{v} : t \in \mathbb{R}\}$ orthogonal to the plane with Cartesian equation $x + y + 2z = -5$? You should justify your answer.

Solution: The vector parallel to the line is $\mathbf{v}$. The line is orthogonal to the plane if and only if $\mathbf{v}$ is parallel to the normal vector of the plane. But the normal vector of the plane is $\mathbf{n} = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$. Since $\mathbf{v}$ and $\mathbf{n}$ are not parellel, the answer is no.

5. **[3 points]** Determine all values of the parameter h such that the linear system

$$\begin{cases} x_1 - x_2 + 2hx_3 & = -1 \\ -3x_1 + 3x_2 + 4x_3 & = 5 \end{cases}$$

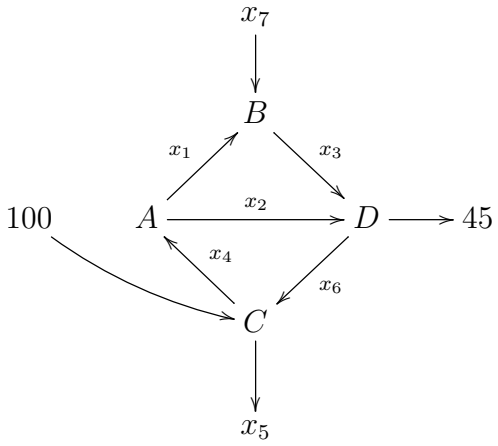
is inconsistent. You should justify your answer.

Solution: We write the augmented matrix of this system and do row reduction towards an E.F. as follows

$$\left[\begin{array}{ccc|c} 1 & -1 & 2h & -1 \\ -3 & 3 & 4 & 5 \end{array}\right] \xrightarrow{R_2 \to R_2 + 3R_1} \left[\begin{array}{ccc|c} 1 & -1 & 2h & -1 \\ 0 & 0 & 4+6h & 2 \end{array}\right]$$

The system is inconsistent if and only if $4 + 6h = 0$, that is, when $h = -\dfrac{2}{3}$.

6. Consider the traffic flow described by the following diagram. The letters A through D label intersections. The arrows indicate the direction of flow (all roads are one-way) and their labels indicate flow in cars per hour.



(a) **[2 points]** Write down a linear system describing the traffic flow, i.e., all constraints on the variables $x_i, i = 1, \ldots, 7$. (You do not need to solve the linear system.) **Solution:**

$$\begin{cases} A: & x_4 = x_1 + x_2 \\ B: & x_7 + x_1 = x_3 \\ C: & 100 + x_6 = x_4 + x_5 \\ D: & x_2 + x_3 = 45 + x_6 \\ \text{total}: & 100 + x_7 = 45 + x_5 \end{cases}$$

(b) **[1 point]** The reduced echelon form of the augmented matrix of this linear system is as follows:

$$\left[\begin{array}{ccccccc|c} 1 & 0 & -1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & -1 & 0 & 45 \\ 0 & 0 & 0 & 1 & 0 & -1 & 1 & 45 \\ 0 & 0 & 0 & 0 & 1 & 0 & -1 & 55 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right].$$

Write down the general solution of this system.

Solution:

$$\begin{aligned} x_1 &= x_3 - x_7, \\ x_2 &= 45 - x_3 + x_6, \\ x_3 &= \text{free}, \\ x_4 &= 45 + x_6 - x_7, \\ x_5 &= 55 + x_7, \\ x_6 &= \text{free}, \\ x_7 &= \text{free}. \end{aligned}$$

(c) [**1 point**] Suppose that the road x_6 is closed. What is the maximum number of cars per hour that can pass through the road x_4?

Solution: We have $x_6 = 0$, so $x_4 = 45 - x_7 \leq 45$ because $x_7 \geq 0$.